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FDACNet: Enhancing time-series classification with fuzzy feature and integrated self-attention and temporal convolution

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ABSTRACT

Time-series classification is crucial in time series analysis and holds significant importance in real-world scenarios. Applying self-attention and temporal convolution techniques is paramount when dealing with time series data. The self-attention mechanism enables the capture of correlations between different time steps in a sequence, thereby facilitating the handling of long-term dependencies. Meanwhile, temporal convolution is designed explicitly for processing time series data, effectively capturing temporal dependencies through convolutional layers. The integration of the two technologies plays a pivotal role in time series analysis, enabling accurate temporal classification. This paper proposes a novel net with fuzzy features and integrated self-attention and temporal convolution, denoted as FDACNet. The proposed net introduces two key components: FD-FE for fuzzy dominated feature extraction, and ATCmix for integrating self-attention and temporal convolution. FD-FE captures trend information by defining gradient relationship between time points within a time series sample. On the other hand, ATCmix combines convolution and self-attention to reduce parameters and enhance efficiency in handling time-series data. Finally, the proposed method is evaluated on twenty datasets and compared against twelve other state-of-the-art approaches. Experimental results demonstrate the superior classification accuracy of the proposed model, showcasing a 5.2% and 7.1% enhancement in average accuracy compared to the state-of-the-art convolution-based and transformer-based methods ModernTCN and iTransformer.

1. Introduction

Time series data, which consists of a sequence of observations collected over time, is widely encountered in diverse real-life applications such as finance [1], medical [2], and anomaly detection [3]. Time series data analysis has emerged as a crucial research area due to its potential to extract valuable insights and enhance decision-making processes. Time series analysis techniques are vital in dealing with time-series data, with time-series classification (TSC) emerging as a pivotal task across a wide array of real-world applications [4]. TSC can play a crucial role in equipment condition monitoring and operational state assessment, where accurate classification of time-series data facilitates early fault detection, predictive maintenance, and optimized resource allocation. For instance, for nuclear power plants, parameters such as temperature, pressure, steam flow rate, and feedwater flow rate can be monitored continuously over time using sensors. By constructing a TSC model, these data can be utilized for operational condition monitoring. Subsequently, in the event of abnormal conditions, prompt maintenance actions can be taken to mitigate the risk of further losses.

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In daily life, where uncertainty abounds, addressing the inherent uncertainty in time-series data has become crucial [5–7]. Fuzzy set theory, proposed by Zadeh [8], has emerged as a powerful tool for this purpose. Fuzzy sets enable the modeling of membership degrees, effectively capturing vague trends and patterns. By allowing for degrees of membership rather than strict binary distinctions, fuzzy sets provide a more nuanced representation of complex and uncertain data, enhancing our ability to analyze and interpret information that does not fit neatly into traditional categories. Building on the fuzzy set theory, this paper proposes a fuzzy dominated feature extraction block, denoted as FD-FE. By defining gradient relationship between time points and constructing a fuzzy dominated matrix, FD-FE quantifies the dominated strength among temporal samples, thereby enhancing the interpretability and robustness of feature extraction in noisy or non-stationary environments.

Meanwhile, advancements in deep learning have revolutionized time-series analysis. In recent years, significant advancements have been made in time series analysis by applying self-attention mechanisms and temporal convolution. Self-attention mechanisms, exemplified by models such as the Transformer [9], Informer [10], and Autoformer [11], excel in capturing long-term dependencies by dynamically allocating weights across different time steps, thereby addressing the gradient vanishing issues inherent in traditional RNN [12] and LSTM [13]. These mechanisms have proven particularly effective in modeling complex temporal relationship, enhancing prediction accuracy through multi-head attention, and optimizing computational strategies such as sparse attention and auto-correlation.

On the other hand, temporal convolution [14] and its enhanced models [15,16], with their unique architecture that combines causal and dilated convolutions, offer high computational efficiency and parallelizable training, effectively capturing local patterns while expanding the receptive field without losing temporal sequencing. Despite their strengths, conventional temporal convolution is limited by their restricted effective receptive fields (ERFs), which hinder their ability to model long-term dependencies as effectively as Transformer-based models. To bridge this gap, recent innovations, such as ModernTCN [17], have introduced modern convolutional designs, including depthwise separable and pointwise convolutions, to expand ERFs and improve long-term dependency modeling significantly. The integration of self-attention and temporal convolution aims to synergize their complementary strengths: self-attention's ability to model global dependencies and temporal convolution's efficiency in capturing local patterns. As exemplified by the proposed ATCmix block, this hybrid approach reduces parameter redundancy, enhances computational efficiency, and improves the overall robustness of time series models, making it a promising solution for complex temporal analysis tasks.

The motivation for this study arises from two primary challenges encountered in current TSC methods:

(1) Handling Temporal Uncertainty: Time series data often exhibit inherent volatility and are susceptible to noise interference during data collection, posing difficulties for traditional classification techniques in capturing their subtle trends effectively. It is critical to devise a feature extraction mechanism that can adeptly manage temporal uncertainty.

(2) Balancing long-term Dependency Modeling and Computational Efficiency: Current time series models face the challenge of reconciling the modeling of long-term dependencies with preserving computational efficiency. While self-attention mechanisms excel at handling long-term dependencies, they come with high computational complexity. In contrast, traditional TCNs are known for their high efficiency but are limited by their ERFs. This trade-off between complexity and efficiency underscores the critical need for hybrid solutions that can leverage the strengths of both approaches while mitigating their respective limitations.

To tackle these challenges, this paper makes three significant contributions:

1) Uncertainty Modeling: This paper establishes a novel fuzzy dominated relationship to extract trend information from time series data by leveraging fuzzy set theory. The subsequent utilization of the relationship matrix as a feature for network training yields robust interpretability in feature extraction.

2) Hybrid Architecture Optimization: This study integrates temporal convolution with self-attention mechanisms to craft a parameter-efficient hybrid block ATCmix. This integration synergistically enhances the model's capacity to balance long-term dependency modeling and computational efficiency.

The structure of this article is as follows: Section 2 introduces the fundamental concepts of fuzzy set theory and fuzzy dominated relationship, and reviews related work on self-attention and temporal convolution. Section 3 presents the proposed fuzzy dominated extraction method, FD-FE, and the fusion module ATCmix, which integrates self-attention and temporal convolution. Finally, in Section 4, testing is conducted on 20 datasets, comparing the proposed method with state-of-the-art TCN and Transformer architectures. The results demonstrate the superiority of the proposed approach.

2. Related work

This section introduces the basic concept of the fuzzy dominated relation in fuzzy set theory, and offers an overview of the self-attention and temporal convolution in time-series analysis.

2.1. Fuzzy dominated relationship

In fuzzy set theory, the division of sample space and extraction of spatial information are based on the fuzzy relationship among samples. For any feature p , the fuzzy dominated relation can be defined as [18]

$$R_p^<(x_i, x_j) = \frac{1}{1 + e^{-k \times (v(x_i, p) - v(x_j, p))}}, \quad (1)$$

where k is a positive integer, and $v(x, p)$ denotes the value of sample x under feature p .

The fuzzy dominated relation plays a crucial role in time analysis. Through fuzzy modeling, it exhibits flexibility in handling noise and uncertainty within time-series data, aiding in capturing ambiguous trends and extracting meaningful information.

2.2. Self-attention in time-series analysis

In recent years, the application of self-attention mechanisms in time series analysis has made significant advancements, emerging as a pivotal technology for addressing complex temporal modeling challenges [10,11,19–21]. Self-attention mechanisms are highly valuable in time series processing as they can capture correlations between different time steps within a sequence, effectively handling long-term dependencies. By employing dynamic weight allocation, self-attention mechanisms efficiently capture long-term dependencies in time series, thereby mitigating the gradient vanishing issues encountered by traditional RNN [12] and LSTM [13] when dealing with lengthy sequences. For instance, the Transformer, through its multi-head attention mechanism, has successfully modeled relationship at arbitrary positions within a sequence, notably enhancing the accuracy of time series predictions. Recent research has further optimized the computational efficiency of self-attention mechanisms; for instance, models like Informer [10] and Autoformer [11] have reduced computational complexity while maintaining model performance by introducing sparse attention and auto-correlation mechanisms.

2.3. Temporal convolution in time-series analysis

In recent years, significant progress has been made in the research of convolution in time series analysis, with particular attention focused on TCN [14] and its enhanced models [15,16] for complex temporal modeling. TCN, distinguished by its unique convolutional architecture, effectively captures local patterns and long-term dependencies within time series data, offering advantages in high computational efficiency and ease of parallelizable training. Compared to traditional RNN and LSTM, TCN circumvents the issue of gradient vanishing. By combining Causal Convolution and Dilated Convolution, TCN expands its receptive field without losing temporal sequencing, thereby enhancing its ability to model long sequential data. However, conventional TCNs face limitations in capturing long-term dependencies due to their restricted ERFs, resulting in performance that lags behind that of Transformer and MLP models. To address this, ModernTCN [17] reconstructs the traditional TCN by integrating modern convolutional designs such as depthwise separable convolutions and pointwise convolutions. By incorporating large convolutional kernels to significantly expand the ERF, ModernTCN excels in modeling long-term dependencies, positioning itself as a standout performer in long-term dependency modeling within time series analysis.

3. The proposed method

3.1. FD-FE: the fuzzy dominated feature extraction

Time-series data often exhibit inherent volatility and susceptibility to noise interference during data collection, which poses challenges for traditional classification methodologies in effectively capturing nuanced trends. There is a pressing need to develop a feature extraction mechanism capable of handling temporal uncertainty. The adaptability of fuzzy dominated relations in managing noise and uncertainty within time-series data enables the capture of ambiguous trends and the extraction of meaningful insights. Subsequently, a novel fuzzy dominated relation is defined for time-series data to extract intrinsic trend information precisely.

Given a time-series sample $X \in R^{M \times L}$, for any time point t_s , the gradient of any channel c_m is defined as

$$g(t_s, c_m) = \begin{cases} \frac{1}{2} (v(t_{s+1}, c_m) - v(t_s, c_m)), & 1 < s < L, \\ v(t_{s+1}, c_m) - v(t_s, c_m), & \text{else,} \end{cases} \quad (2)$$

where $v(t_s, c_m)$ denote the value of t_s under the channel c_m .

Given a time-series sample $X \in R^{M \times L}$, the fuzzy dominated relationship feature matrix is defined as

$$S(X) = [S_{ij}]_{L \times L}, \quad (3)$$

where $S_{ij} = \frac{1}{1 + e^{-k \times \sqrt{\sum_m (sign(g(t_i, c_m)) - sign(g(t_j, c_m))))^2}}$, k is a positive real number, and $sign(x)$ denote the sign function.

Furthermore, the proposed fuzzy dominated relation S_{ij} has the following proposition:

$$\frac{1}{2} \leq S_{ij} \leq \frac{1}{1 + e^{-2k\sqrt{m}}}. \quad (4)$$

The proposition can be easily demonstrated because $0 \leq |sign(g(t_i, c_m)) - sign(g(t_j, c_m))| \leq 2, \forall i, m$. It can be found that when $m \rightarrow \infty$, we have $\frac{1}{1 + e^{-2k\sqrt{m}}} \rightarrow 1$. Therefore, concerning the volatility and noise interference present in time-series data, on the one hand, the stable points in the feature relation matrix $[S_{ij}]_{L \times L}$ remain unaffected. On the other hand, the fuzzy relation values of points affected also exhibit minimal variations, typically within the range of $[1/2, 1]$. The fewer channels there are, the less interference is encountered. In general, the values in the fuzzy dominated relationship feature matrix indicate the proximity of the increasing or decreasing trends between any two time points in a time series; the larger the value, the greater the disparity in trend between those two points. Therefore, the proposed dominated relationship among time points can reflect the overall trend of time-series samples, thereby aiding in modeling Time Series Classification (TSC). Fig. 1 showcases the FD-FE features of two classes of samples

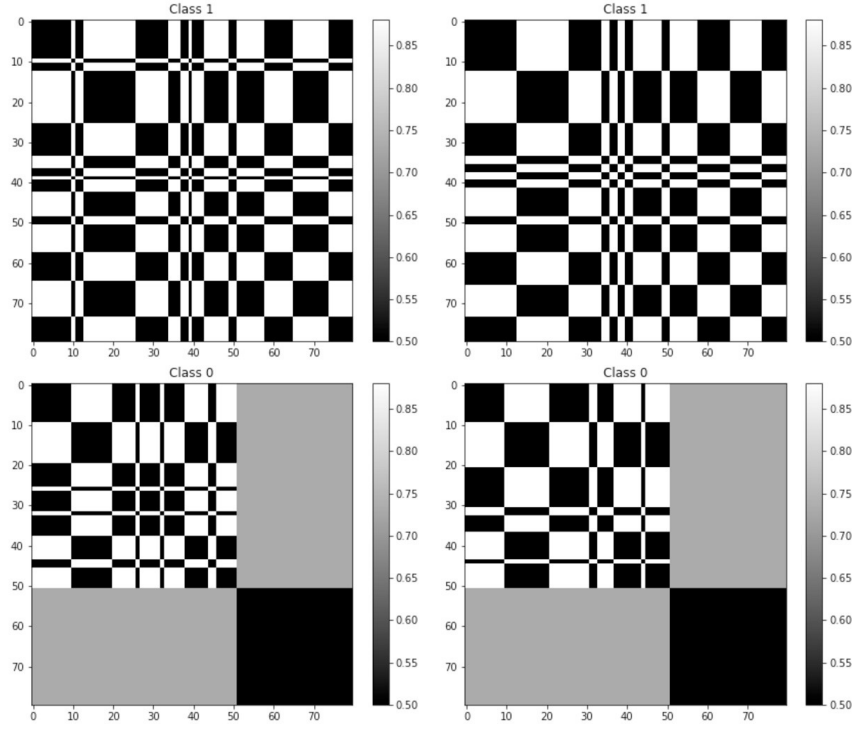


Fig. 1. Illustration of FD-FE features in PhalangesOutlinesCorrect. Similar samples exhibit close FD-FE characteristics, while distinct samples show significant differences.

in the PhalangesOutlinesCorrect dataset. Similar samples exhibit close FD-FE characteristics, while distinct samples show significant differences.

The parameter k controls the steepness of the fuzzy relationship, determining how quickly distance influences the relationship within the model. A higher k value increases sensitivity to small distance variations, resulting in a sharper decision boundary (escalating swiftly from 0.5 to 1). Once the distance surpasses a threshold, it is promptly recognized as a significant dissimilarity. With an elevated k , dominated relationship between samples becomes highly responsive to symbol dissimilarity distances, leading to distinct boundaries. Conversely, a lower k yields gentler transitions in dominated relationship, even when symbol dissimilarities are substantial, implying some degree of similarity. This principle is particularly useful in scenarios necessitating fuzzy treatment, such as handling noisy or uncertain data. Typically, the commonly used k value is set to 1 [22]. In practical applications, the k value can be adjusted based on specific requirements. Table 8 provides insights into how various k values impact model performance.

3.2. ATCmix: the integration of self-attention and temporal convolution

Inspired by [23], convolution and self-attention are beneficial for extracting relationship between data. Combining the two can reduce model parameters and enhance model efficiency. To better handle time-series data, we adopt Temporal Convolutional Network (TCN) [14]. The specific derivation steps are as follows:

3.2.1. The separated stages of temporal convolution

TCN is a neural network architecture designed specifically for processing sequential data. It utilizes one-dimensional convolutional layers to capture temporal dependencies within the data effectively. By stacking multiple dilated causal convolutional layers, TCN can efficiently model local dependencies. Next, we can decompose the standard TCN into two steps: convolution and shift. For simplicity, let's assume a stride of 1.

For a standard dilated causal convolution kernel $K \in R^{D_{out} \times D_{in} \times k}$ with dilation factor d , where D_{in} and D_{out} denote the dimensions of input and output tensors, and k is the kernel size. Let $X_{in} \in R^{D_{in} \times L}$ and $X_{out} \in R^{D_{out} \times L}$ be the input and output tensors, where L is the length of series. Assume that $x_{in}(i) \in R^{D_{in}}$ and $x_{out}(i) \in R^{D_{out}}$ are the time point i of X_{in} and X_{out} . Then, the output time point can be rewritten as follows:

$$x_{out}(i) = K_0 x_{in}(i - (k-1)d) + K_1 x_{in}(i - (k-2)d) + \dots + K_{k-1} x_{in}(i), \quad (5)$$

where $K_p \in R^{D_{out} \times D_{in}}$, $p = 1, 2, \dots, k-1$, denotes the kernel weights of the position p .

Now, we define the shift operation as $S(x_{in}(i), \Delta i) = x_{in}(i + \Delta i)$. Thus, the formula (5) is equivalent to

$$\begin{aligned} x_{out}(i) &= K_0 S(x_{in}(i), -(k-1)d) + K_1 S(x_{in}(i), -(k-2)d) + \dots + K_{k-1} S(x_{in}(i), 0) \\ &= S(K_0 x_{in}(i), -(k-1)d) + S(K_1 x_{in}(i), -(k-2)d) + \dots + S(K_{k-1} x_{in}(i), 0) \end{aligned} \quad (6)$$

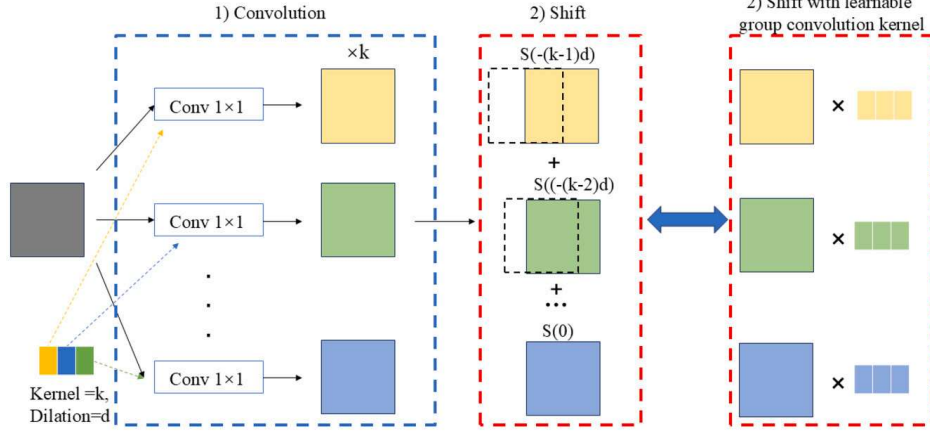


Fig. 2. Illustration of the separated temporal convolution. Firstly, calculate the convolution result for k parts of the kernel, then perform a shifting operation, and finally add the results together. Moreover, for operational convenience, the shift operation can be designed as a convolution operation, and the parameters within the convolutional kernel can also be learned.

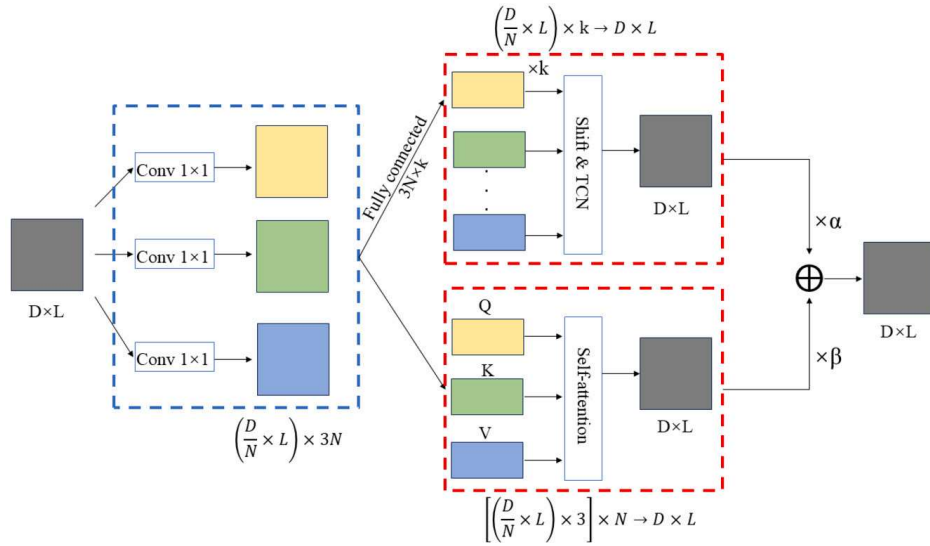


Fig. 3. Illustration of the proposed ATCMix. Initially, the input tensor is projected into three projection matrices using 1×1 convolutional kernels, which can be treated as Q, K, V, and fed into the self-attention component for weighting. Simultaneously, the three projection matrices are mapped to k matrices using fully connected layers, where k represents the kernel size. Subsequently, a shift with learnable group convolution kernels is applied to these k matrices. After that, a TCN block is employed to extract more temporal features from the obtained tensor.

Consequently, the standard dilated causal convolution can be decomposing to two steps:

1) Convolution: $K_p x_{in}(i)$, $p = 1, 2, \dots, k - 1$;

2) Shift: $S(K_0 x_{in}(i), -(k-1)d) + S(K_1 x_{in}(i), -(k-2)d) + \dots + S(K_{k-1} x_{in}(i), 0)$.

According to the derivation, consistent with [23], we can decompose the convolution kernel into two parts when computing temporal convolutions. Firstly, calculate the convolution result for k parts of the kernel, then perform a shifting operation, and finally add the results together. Moreover, for operational convenience, the shift operation can be designed as a convolution operation, and the parameters within the convolutional kernel can also be learned. The specific illustration can be seen in Fig. 2.

3.2.2. The integration of self-attention and temporal convolution

Self-attention can be clearly divided into two parts: projection first, followed by weighted summation. Therefore, similar to the structure in [23], we integrate temporal convolution and self-attention to reduce training parameters and enhance model efficiency. Initially, the input tensor is projected into three projection matrices using 1×1 convolutional kernels, which can be treated as Q, K, V, and fed into the self-attention component for weighting. Simultaneously, the three projection matrices are mapped to k matrices using fully connected layers, where k represents the kernel size. Subsequently, a shift with learnable group convolution kernels is applied to these k matrices. After that, a TCN block is employed to extract more temporal features from the obtained tensor. The architecture is shown in Fig. 3.

The integration of self-attention and temporal convolution in the ATCMix model enhances the effective capture of long-term and local dependencies within time series data. This is achieved by representing features at multiple scales and learning hierarchical representations. The self-attention mechanism learns global relationship for hierarchical representation, while temporal convolution

extracts local patterns. Combining these mechanisms allows the model to learn features at various scales, effectively capturing both long-term dependencies and local details in the data.

Furthermore, self-attention and temporal convolution operate within distinct receptive fields, with self-attention focusing on broader global information and temporal convolution on extracting local details. This difference in receptive fields significantly contributes to the model's comprehensive understanding of time series data.

In summary, the ATCmix model combines the strengths of self-attention and TCN, allowing for the concurrent capture of global and local dependencies with a broad receptive field. This enhances the model's ability to learn representations from time series data. Self-attention grasps global information, while TCN excels in extracting local patterns. Through this integration, ATCmix achieves a more holistic comprehension of intricate patterns and dynamic variations in time series data.

3.3. Model architecture

This section delineates the overall architecture of the FDACNet, encompassing: Embedding, ATCmix block, FD-FE block, Backbone, and Output.

Embedding: For the input tensor $X_{in} \in R^{D_{in} \times L}$, we utilize the 1D convolution to obtain an embedding tensor $X_{emb} \in R^{D_{out} \times L}$. Considering the scenario where the kernel size is larger than the stride, to avoid mixing information from subsequent time points, we pad the front of the sequence with $p - s$ zeros, where p and s denote the kernel size and stride of the embedding layer. In this paper, the parameters p and s are set to be 1 by default.

ATCmix block: The ATCmix block unfolds with the ATCmix module initiating feature extraction, enabling the model to grasp correlations and patterns within the data effectively. Subsequently, Batch Normalization1d is applied to accelerate training, enhance model generalization, and prevent issues with gradient vanishing or exploding. Following this, the one-dimensional ConvFFN network processes the features, further abstracting them to capture intricate data patterns and features accurately. Finally, the block incorporates residual connections, where the processed features are combined with the input data, thereby enhancing the flow of information and facilitating gradient propagation. The ATCmix block can be formulated as:

$$X_{ATCmix} = l \times \{ConvFFN(bn(ATCmix(X_{emb}))) + X_{emb}\}, \quad (7)$$

where l denotes the layers of the ATCmix block.

FD-EF block: The FD-FE module is employed to extract the fuzzy dominated relationship feature matrix. Then it is fed into the ACmix module [23] to further kickstart the feature extraction process, aiding the model in comprehending data correlations and patterns efficiently. Additionally, Batch Normalization1d is applied to standardize input distributions across intermediate layers, enhancing training speed and stability. The subsequent two-dimensional convolution (Conv2d) processes the features, enabling non-linear transformations for extracting more intricate feature representations. Finally, the block also incorporates residual connections, combining the processed features with the input data. The FD-EF block can be formulated as follows:

$$X_{FD-FE} = l \times \{Conv2d(bn(ACmix(FD - FE(X_{in})))) + X_{in}\}, \quad (8)$$

where l denotes the layers of the FD-EF block.

Backbone: For the embedding tensor $X_{emb} \in R^{D_{out} \times L}$, the ATCmix block is utilized to extract features. In addition, for the input tensor $X_{in} \in R^{D_{in} \times L}$, the FD-FE is employed to extract the fuzzy dominated feature matrix directly, and then it is fed into the ACmix block. Subsequently, the features from both processes are weighted to obtain the final feature, which is formulated as:

$$X_{Backbone} = \varphi X_{ATCmix} + (1 - \varphi) X_{FD-FE}, \quad (9)$$

where $\varphi \in [0, 1]$ denotes the threshold.

Output: After passing through the Backbone, the $X_{Backbone} \in D \times L$ is first flattened to $X_{Backbone} \in 1 \times (D \times L)$, then a one-dimensional linear Head and softmax activation function are used to obtain the final classification result $X_{Output} \in 1 \times c$, where c represents the number of classes.

4. Experimental analysis

This section conducts a series of experiments to showcase the proposed model's effectiveness, all executed on a system equipped with PyTorch 2.0.0 and an NVIDIA Quadro RTX4000 8G GPU. We sourced twenty datasets from the public time-series database.¹ These datasets cover diverse domains, including human activity recognition (HAR), audio recognition, EEG signal recognition, traffic flow forecasting, and weather monitoring, detailed in Table 1.

4.1. Baselines

Our model is compared with 10 state-of-the-art deep learning and 2 classical models. The details of these baselines are as follows:

¹ www.timeseriesclassification.com.

Table 1
Details of the datasets.

No.	Datasets	Train size	Test size	Dimensions	Length	Classes	Domain
1	ERing	30	270	4	65	6	HAR
2	BasicMotions	40	40	6	100	4	HAR
3	EyesOpenShut	56	42	14	128	2	EEG
4	DuckDuckGeese	60	40	1345	270	5	AUDIO
5	Lightning2	60	61	1	637	2	Weather
6	Cricket	108	72	6	1197	12	HAR
7	RacketSports	151	152	6	30	4	HAR
8	NATOPS	180	180	24	51	6	HAR
9	ArticularyWordRecognition	275	300	9	144	25	MOTION
10	LargeKitchenAppliances	375	375	1	720	3	IoT
11	EthanolLevel	504	500	1	1751	4	SPECTRO
12	MindReading	727	653	204	200	5	MEG
13	EMOPain	1093	50	30	180	3	HAR
14	MelbournePedestrian	1194	2439	1	24	10	TRAFFIC
15	PhalangesOutlinesCorrect	1800	858	1	80	2	IMAGE
16	LSST	2459	2466	6	36	14	OTHER
17	Crop	7200	16800	1	46	24	IMAGE
18	PenDigits	7494	3498	2	8	10	MOTION
19	ElectricDevices	8926	7711	1	96	7	DEVICE
20	Tiselac	81714	17973	10	23	9	IMAGE

1) ModernTCN (2024): [17] ModernTCN introduces advanced convolutional designs like depthwise separable and pointwise convolutions, greatly enhancing ERFs and refining long-term dependency modeling.

2) TimesNet (2023): [24] TimesNet revolutionizes the process by mapping one-dimensional time series data onto a two-dimensional spatial representation, enabling the modeling of both intra-periodic and interperiodic information. This enhancement significantly boosts its ability to grasp intricate temporal characteristics.

3) Informer (2020): [10] Informer enhances the Transformer with efficient ProbSparse self-attention, self-attention distillation, and parallel generative decoding mechanisms, improving computational efficiency and prediction capabilities for long sequences.

4) PatchTST (2023): [25] PatchTST is a novel time series forecasting model that leverages patch-based processing and Transformer architecture to capture temporal dependencies and enhance prediction accuracy effectively.

5) Autoformer (2021): [11] Autoformer is a Transformer-based model for time series forecasting, utilizing a novel decomposition architecture to capture long-term dependencies and seasonal patterns through an auto-correlation mechanism.

6) Nonstationary Transformer (2022): [21] The Nonstationary Transformer effectively models nonstationary time series by incorporating adaptive mechanisms that dynamically adjust to changing statistical properties over time, enabling improved capture of local and global dependencies, especially beneficial for real-world datasets with irregular patterns like financial or environmental data.

7) Crossformer (2023): [19] Crossformer employs the Dimension Segment-Wise approach to encode the input into a two-dimensional vector, ensuring the retention of both temporal and dimensional details. Additionally, it incorporates a dual-stage attention mechanism to capture dependencies across time and dimensions efficiently.

8) iTransformer (2024): [20] iTransformer individually processes distinct variables, encoding each as an autonomous token. Leveraging attention mechanisms, it captures inter-variable correlations, while employing feed-forward networks to model temporal relationship among variables. These strategies enhance its ability to acquire superior temporal representations.

9) DLinear (2023): [26] DLinear confirms that a channel-independent simple MLP surpasses the conventional Transformer model, sparking considerations regarding Transformers and channel independence.

10) LightTS (2022): [27] LightTS is a lightweight and efficient time series forecasting model that prioritizes reduced computational complexity alongside high performance, utilizing simplified architectures and optimization techniques to ensure fast training and inference times, making it ideal for resource-constrained environments and real-time forecasting applications.

11) TimeSeriesForest (2013): [28] TimeSeriesForest is a tree-ensemble method for time series classification that leverages Entrance gain, a combination of entropy gain and distance measure, to evaluate splits, showcasing improved accuracy with computational efficiency and a temporal importance curve capturing useful temporal characteristics for classification, outperforming competitors like dynamic time warping-based classifiers.

12) LSTM (1997): [13] LSTM is a classic recurrent neural network variant designed to capture long-term dependencies in sequential data. It uses memory cells and gating mechanisms to retain information over extended periods, making it effective for time series forecasting tasks.

4.2. Implementation details

All models are trained using cross-entropy loss and optimized with ADAM [29]. To prevent overfitting, an early stopping training strategy with a patience of 10 is employed, while the number of epochs is fixed at 30 for efficiency. The learning rate is fine-tuned within the range of $\{0.0005, 0.001\}$, and the batch size within $\{32, 64\}$. The default output dimension of the embedding layer is set

Table 2

Comparison results of classification accuracy. A P-value falling below the significance level thresholds signifies the significant outperformance of our method relative to the others.

Datasets	Ours	Convolution-based		Transformer-based						MLP-based		Classical methods	RNN-based
		ModernTCN	TimesNet	Informer	PatchTST	Autoformer	Nonstationary Transformer	Crossformer	iTransformer	DLinear	LightTS	TimeSeriesForest	LSTM
PenDigits	97.6	97.6	98.1	96.3	95.6	96.4	98.0	91.8	94.9	88.6	95.8	89.5	82.8
BasicMotions	91.0	90.5	96.5	96.0	61.0	54.0	99.5	73.0	91.0	72.5	57.0	98.5	42.0
ElectricDevices	69.3	66.0	69.3	64.4	63.1	58.5	70.8	60.8	56.9	47.4	58.0	63.2	56.4
LSST	38.6	33.8	36.8	35.9	51.2	41.4	38.4	35.8	36.4	31.7	36.6	43.0	31.5
Crop	74.0	72.2	74.6	71.2	71.2	68.9	72.3	56.1	66.5	67.2	70.0	65.6	16.6
MelbournePedestrian	93.2	84.8	94.6	90.6	84.8	80.3	87.4	82.3	86.0	59.2	84.5	91.5	46.9
PhalangesOutlinesCorrect	77.1	65.5	71.0	65.5	64.5	67.1	66.2	63.8	65.1	62.6	68.2	74.5	61.3
EthanolLevel	50.9	29.0	28.2	28.9	28.2	33.3	28.0	26.4	26.0	29.3	34.5	42.7	25.0
Cricket	96.4	92.5	94.4	91.1	93.3	24.7	92.5	75.3	90.8	90.3	88.9	83.9	7.2
EMOPain	84.3	81.9	83.1	85.5	81.2	70.5	82.7	78.5	84.1	78.3	79.7	78.2	69.4
NATOPS	94.3	84.8	87.4	91.6	63.9	79.2	91.3	84.8	80.1	89.0	83.7	80.4	18.0
RacketSports	82.5	74.1	84.9	86.6	73.2	78.6	87.1	78.9	79.2	66.8	70.5	74.5	25.4
ArticularyWordRecognition	96.5	97.8	97.7	98.0	82.5	58.5	95.3	96.9	98.3	96.9	61.6	87.7	6.6
Tiselac	91.5	92.2	92.5	82.2	80.2	77.6	92.1	22.3	77.2	82.5	89.4	86.8	80.6
MindReading	44.5	50.8	43.2	41.6	25.3	28.4	42.8	41.7	38.9	39.1	41.7	23.1	21.1
DuckDuckGeese	61.6	49.6	58.4	55.6	21.6	29.2	54.8	29.6	46.4	62.8	51.6	38.8	24.0
ERing	94.4	89.9	91.3	93.7	92.4	73.0	93.8	86.1	93.6	70.6	64.3	77.8	16.2
LargeKitchenAppliances	60.3	44.0	57.4	52.4	66.1	42.9	63.0	52.2	44.7	44.5	43.7	54.0	34.2
Lightning2	74.1	72.5	75.4	75.7	69.5	55.1	77.7	71.1	73.4	71.8	73.4	77.4	52.8
EyesOpenShut	57.6	56.2	57.1	62.4	55.7	54.8	61.9	56.7	57.6	63.3	61.4	53.8	51.9
Average	76.5	71.3	74.6	73.3	66.2	58.6	74.8	63.2	69.4	65.7	65.7	69.2	38.5
Average ranking	11.05	7.83	10.40	9.30	5.98	4.60	10.28	5.03	6.53	5.53	6.25	6.95	1.30
Nemenyi P-values		0.30	1.00	0.97	< 0.01	< 0.001	1.00	< 0.001	< 0.05	< 0.001	< 0.01	< 0.05	< 0.001
Wilcoxon P-values		< 0.001	< 0.1	< 0.05	< 0.01	< 0.001	0.18	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

to 2048, with 2 layers and a kernel size of 13 for ACmix and ATCmix models. Parameters α , β , and φ are uniformly set to 0.5. Strides within {10, 15, 20} in the FD-FE block are fine-tuned for datasets with longer lengths, such as EMOPain, Cricket, and DuckDuckGeese, to downsample and improve computational efficiency. Each model undergoes five tests with different random seeds (2021, 2022, 2023, 2024, 2025), and the average classification accuracy is calculated as the performance metric, as detailed in Table 2.

4.3. Main results

In Table 2, our model demonstrates exceptional performance across a majority of datasets, exhibiting superior classification accuracy. Notably, our model significantly surpasses all other baseline methods in datasets such as PhalangesOutlinesCorrect, EthanolLevel, Cricket, NATOPS, and ERing. Compared to the most advanced convolution-based approaches, ModernTCN and TimesNet, our proposed method outperforms them on datasets such as LSST, EMOPain, and DuckDuckGeese. Moreover, in datasets such as Crop, MelbournePedestrian, and MindReading, our model excels over transformer-based methods including Informer, PatchTST, Autoformer, Nonstationary Transformer, Crossformer, and iTransformer. Additionally, compared to MLP-based and classical methods like DLinear, LightTS, and TimeSeriesForest, our proposed method showcases superior performance in datasets such as PenDigits, ElectricDevices, and RacketSports.

Overall, our method achieves an average accuracy of 76.5, marking a 5.2% and 7.1% enhancement compared to the ModernTCN and iTransformer. This outcome underscores the significant performance edge of our model in time series data classification tasks.

Furthermore, in consideration of inherent randomness, the average classification accuracies are meticulously evaluated using Wilcoxon signed-rank and Nemenyi tests. This statistical analysis is conducted to assess the significance of differences between the proposed FDACNet and the other comparative baselines. Through the calculation of P-values and subsequent comparison with predefined significance levels, we can ascertain whether our method demonstrates statistical superiority over the comparison methods. A P-value falling below the significance level thresholds (0.001, 0.01, 0.05, 0.1) signifies the significant outperformance of our method relative to the others. The detailed outcomes of the Wilcoxon signed-rank and Nemenyi tests are elaborated in Table 2.

The test results reveal that the proposed FDACNet attains an average ranking of 11.05 among the compared baselines. Notably, the P-values derived from the Nemenyi test exhibit statistical significance, indicating that our model performs significantly better than PatchTST, Autoformer, Crossformer, iTransformer, DLinear, LightTS, TimeSeriesForest, and LSTM. Similarly, the Wilcoxon P-values uncover substantial performance disparities, with our model surpassing most algorithms. These findings strongly suggest that our model demonstrates statistically significant superiority over the majority of state-of-the-art algorithms.

To enhance the persuasiveness and interpretability of the results, class activation mapping heatmaps (CAM) are utilized to showcase discriminative classification regions, as illustrated in Fig. 4. This figure displays CAM images of ATCmix and FD-FE for six datasets. It is evident that ATCmix adeptly captures trend features within time series samples, with its discriminative regions encapsulating the essential characteristics of the samples. Moreover, with the exception of ArticularyWordRecognition, FD-FE also exhibits a notable capability in extracting trend features. Furthermore, Fig. 5 offers a detailed depiction of CAM images for two sample classes in the PhalangesOutlinesCorrect dataset, revealing similarities within CAMs of the same class and distinct differences across CAMs of different classes.

4.4. Ablation study

Subsequently, ablation experiments are being carried out to assess the individual contributions of the two proposed blocks, FD-FE and ATCmix, within the classification framework. Without loss of generality, the random seed is set to 2021. To delineate the specific

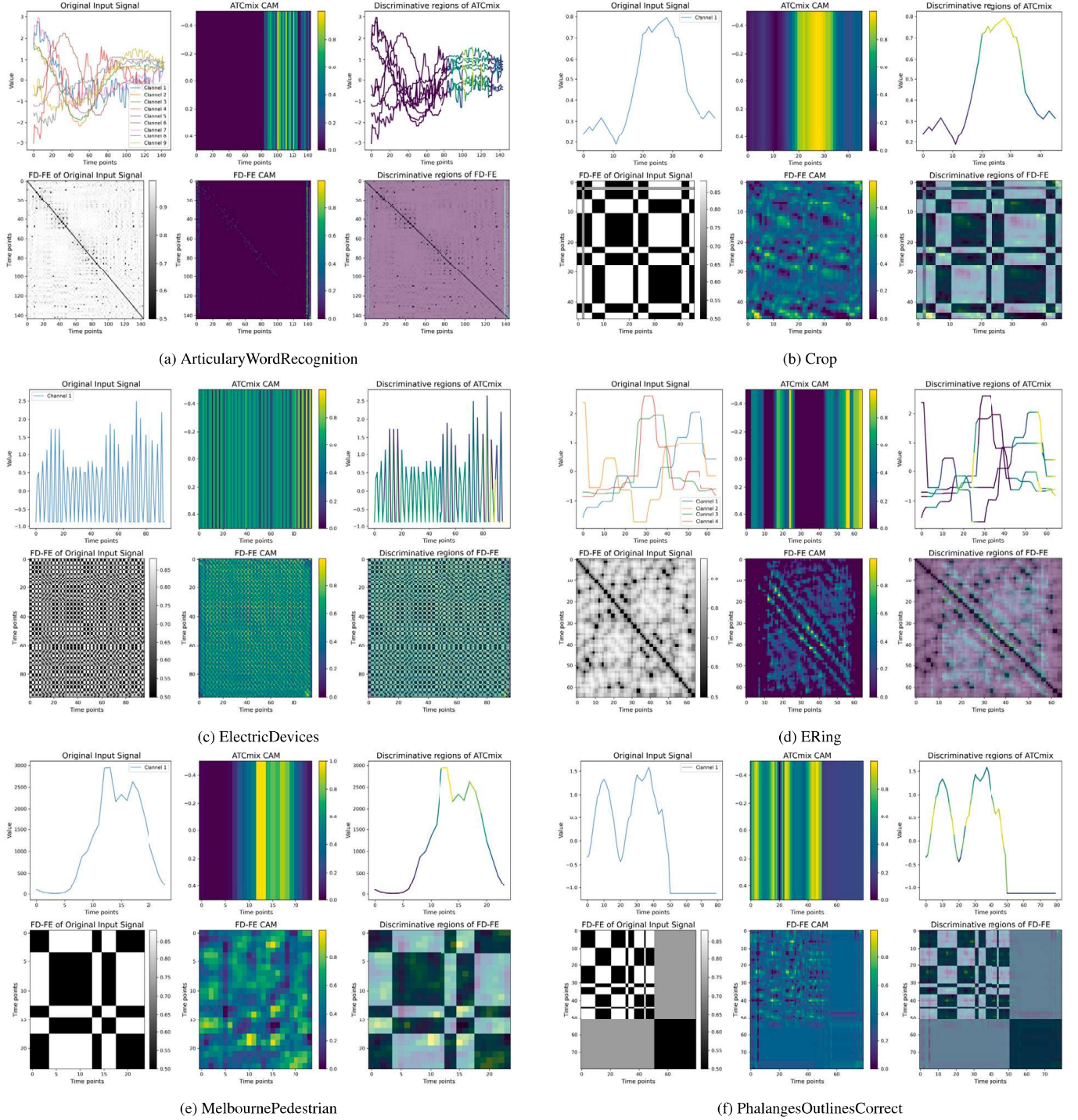


Fig. 4. CAM of ATCmix and FD-FE under six datasets. It is evident that ATCmix adeptly captures trend features within time series samples, with its discriminative regions encapsulating the essential characteristics of the samples. Moreover, with the exception of ArticularWordRecognition, FD-FE also exhibits a notable capability in extracting trend features. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

roles of these blocks, each one is deactivated sequentially, and the resulting classification accuracy is recorded. Furthermore, FD-FE captures trend information by defining gradient relationship between time points within a time series sample. To explore the impact, we substitute the gradient relationship with the classical equivalence relationship (ER) [30] and the dominated relationship (DR) [18] to compare model performance under different relationship settings. These experiments are conducted across twelve datasets, and the ablation results are detailed in Table 3.

The ablation outcomes reveal that the composite model configuration of ATCmix+FD-FE outperforms individual ATCmix and FD-FE across a variety of datasets. Notably, in the BasicMotions dataset, our model attains an accuracy of 90.0%, surpassing the accuracies of ATCmix (77.5%) and FD-FE (47.5%). Similarly, in datasets like PhalangesOutlinesCorrect, EMOPain, NATOPS, Rack-etSports, ArticularWordRecognition, ElectricDevices, LSST, and Crop, our integrated model consistently showcases higher accuracy

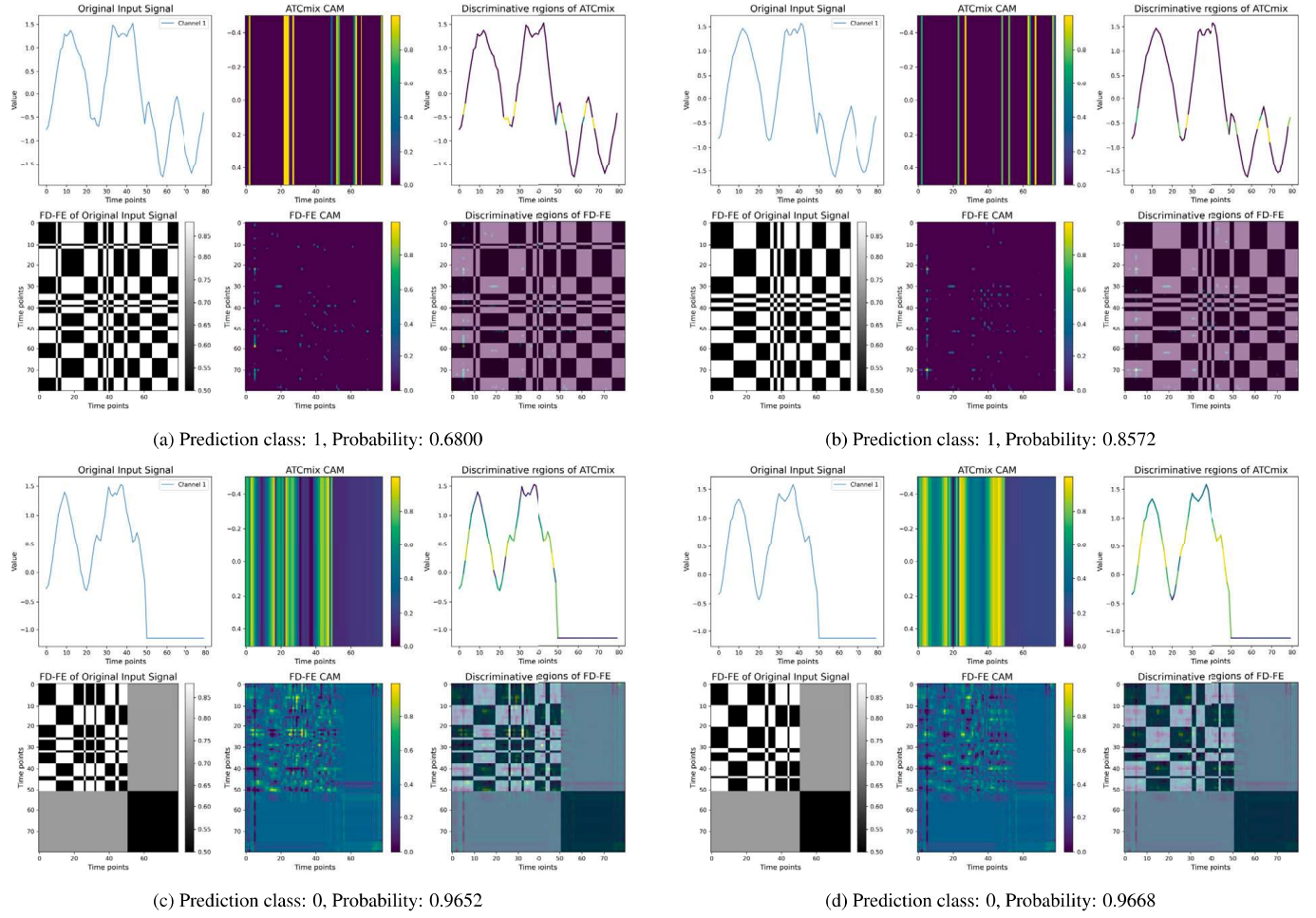


Fig. 5. CAM of ATCmix and FD-FE with respect to different classes in PhalangesOutlinesCorrect datasets. Samples of the same class have similar CAMs and discriminative regions, while those of different categories exhibit significant differences.

Table 3

Ablation results of ATCmix and FD-FE.

Datasets	ATCmix	FD-FE	ATCmix+NR	ATCmix+DR	Ours (ATCmix+FD-FE)
BasicMotions	77.5	47.5	85.0	67.5	90.0
EMOPain	81.4	78.3	81.4	81.4	84.5
NATOPS	93.3	18.9	70.6	74.4	95.0
RacketSports	80.3	30.3	80.3	80.3	82.2
ArticularyWordRecognition	95.3	15.7	97.0	95.3	97.0
PhalangesOutlinesCorrect	74.6	73.3	69.5	68.9	75.5
ERing	93.0	39.6	93.7	93.3	94.4
EyesOpenShut	54.8	50.0	57.1	52.4	57.1
Lightning2	70.5	54.1	54.1	77.0	78.7
LargeKitchenAppliances	59.5	36.5	54.4	57.3	57.1
MelbournePedestrian	93.7	65.2	93.3	85.1	94.3
EthanolLevel	45.4	34.4	41.4	45.0	49.6

levels compared to the individual ATCmix and FD-FE models. And it can be observed that the enhancement effects of NR and DR on ATCmix are generally less pronounced than those of the proposed gradient relationship, indicating the efficacy of the gradient relationship in capturing trend information.

In conclusion, these results underscore the synergistic advantages of merging diverse modules, highlighting the effectiveness of integrating ATCmix and FD-FE components to enhance classification performance across various datasets. The superior performance of our combined model highlights the importance of integrating complementary features from multiple modules to enhance predictive capabilities in classification tasks.

Table 4
Impact of epochs.

Datasets	20	30	50	80	100
BasicMotions	90.0	90.0	90.0	90.0	90.0
EMOPain	84.8	84.8	84.8	84.5	84.8
NATOPS	95.0	95.0	95.0	95.0	95.0
RacketSports	82.2	82.2	82.2	82.2	82.2
ArticularyWordRecognition	97.0	97.0	97.0	97.0	97.0
PhalangesOutlinesCorrect	76.7	75.5	75.9	74.4	75.1
ERing	94.4	94.4	94.4	94.4	94.4
EyesOpenShut	57.1	57.1	57.1	57.1	57.1

Table 5
Impact of kernel size.

Datasets	3	5	7	9	11	13
BasicMotions	97.5	97.5	90.0	80.0	90.0	90.0
EMOPain	83.7	85.4	85.6	87.0	84.5	84.5
NATOPS	96.7	96.1	94.4	96.1	93.9	95.0
RacketSports	82.2	82.2	83.6	83.6	84.2	82.2
ArticularyWordRecognition	97.0	98.0	96.7	96.3	96.7	97.0
PhalangesOutlinesCorrect	77.7	76.8	77.3	76.2	77.0	75.5
ElectricDevices	71.2	71.2	66.8	69.2	70.1	70.1
LSST	39.5	39.1	36.9	38.5	38.6	37.6
Crop	75.0	75.1	73.9	74.3	74.8	73.7

Table 6
Impact of dimensions and layers.

Dimension	Layer	EMOPain	NATOPS	ERing	BasicMotions	EyesOpenShut
1024	2	83.1	93.9	93.3	85.0	52.4
1024	4	83.1	93.9	93.3	85.0	52.4
2048	2	84.5	95.0	94.4	90.0	57.1
2048	4	84.8	95.0	94.4	90.0	57.1
4096	2	82.3	96.1	94.4	95.0	54.8
4096	4	83.1	96.1	94.4	95.0	54.8

Table 7
Impact of strides.

Datasets	10	15	20	25	30
EthanolLevel	49.6	43.2	45.6	40.6	37.8
Cricket	94.4	98.6	93.1	98.6	94.4
EMOPain	84.5	84.5	85.4	83.1	84.8
MindReading	42.6	44.1	42.9	41.8	43.6
DuckDuckGeese	66.0	62.0	66.0	68.0	66.0
LargeKitchenAppliances	57.6	58.7	56.3	57.9	57.1
Lightning2	72.1	78.7	68.9	75.4	72.1

4.5. Parameter sensitivity analysis

Previously, the study fine-tuned the hyperparameters batch size and learning rate, revealing that optimal model performance is typically achieved with a batch size of 32 and a learning rate of 0.001. Subsequently, the impact of varying epochs on model performance is explored, maintaining the batch size at 32 and the learning rate at 0.001, while testing epochs within the {20, 30, 50, 80, 100} range to compare accuracy. Other hyperparameters are also under scrutiny in this subsection, with the random seed set at 2021. Detailed results are provided in Tables 4 - 8.

Upon analyzing Table 4, it becomes evident that different epochs have a marginal effect on model performance, likely due to the early stopping technique implemented. Consequently, setting epoch = 30 in this study is deemed a more rational choice. Moving on to Table 5, distinct accuracies are highlighted across various datasets in response to varying kernel sizes. Datasets like BasicMotions, NATOPS, ArticularyWordRecognition, PhalangesOutlinesCorrect, ElectricDevices, LSST, and Crop demonstrate optimal performance with kernel sizes of 3 or 5, while datasets like EMOPain and RacketSports excel with kernel sizes of 9 and 11. Smaller kernel sizes prove effective in specific datasets, whereas larger kernel sizes show promise in enhancing performance in others. In Table 6, a comparison of model performance across different dimensions and blocks is presented, with dimensions set to 1024, 2048, and 4096, and layers set to 2 and 4. Larger dimensions and layers generally lead to improved model performance, albeit at a higher computational cost.

Table 8
Impact of k in FD-FE.

Datasets	1	3	5	10	100	1000
BasicMotions	90.0	90.0	90.0	90.0	90.0	90.0
EMOPain	84.5	84.2	85.4	85.1	85.1	85.1
NATOPS	95.0	95.0	95.0	95.0	95.0	95.0
RacketSports	82.2	81.6	81.6	81.6	81.6	81.6
ArticularyWordRecognition	97.0	97.0	97.0	97.0	97.0	97.0
PhalangesOutlinesCorrect	75.5	77.6	79.1	76.5	76.8	77.5
ERing	94.4	94.4	94.1	94.1	94.1	94.1
EyesOpenShut	57.1	57.1	57.1	57.1	57.1	57.1
Lightning2	78.7	77.0	77.0	77.0	77.0	77.0
LargeKitchenAppliances	57.1	54.7	56.8	57.6	56.5	57.1
MelbournePedestrian	94.3	93.4	94.0	93.8	94.1	94.3
EthanolLevel	49.6	49.8	50.6	50.6	50.6	50.6

Therefore, based on all considerations, utilizing 2048 dimensions and 2 layers is recommended. Additionally, Table 7 illustrates the impact of different strides on performance, with optimal model performance typically achieved with strides of 10 and 15 for most long sequence datasets. Consequently, strides are adjusted within the range of {10, 15, 20} in this study. Lastly, Table 8 delves into the effect of different values of k in the FD-FE module, highlighting that certain high-dimensional datasets are not sensitive to the value of k . This is because the high dimensionality itself leads to similarity matrix values approaching 1, thereby increasing k does not affect its value. For datasets with lower dimensions, adjusting the value of k proves effective, with optimal values primarily concentrated around {1, 3, 5}. Thus, setting k to 1 by default is a reasonable choice in this study.

All in all, for practical applications, meticulous parameter optimization and dataset-specific experiments are recommended to explore a range of hyperparameters such as kernel sizes, blocks, and strides.

5. Conclusion

In practical applications, on the one hand, time series data often exhibit inherent volatility and are susceptible to noise interference, posing significant challenges for existing models in classification tasks. Furthermore, existing time series models face challenges in balancing the modeling of long-term dependence and computational efficiency. To address these challenges, this study proposes an advanced model that leverages fuzzy set theory for robust feature extraction and integrates self-attention with temporal convolution to enhance both long-term dependency modeling and computational efficiency. The proposed architecture comprises two key components: FD-FE, which employs fuzzy dominated relations for feature extraction, and ATCmix, a hybrid module that combines self-attention and temporal convolution to model long-term dependency while preserving computational efficiency. Extensive experiments on 20 benchmark datasets demonstrate the model's superior performance, achieving an average accuracy improvement of 5.2% and 7.1% over state-of-the-art convolution-based and transformer-based methods ModernTCN and iTransformer. Statistical hypothesis testing further confirms the significance of the proposed model's performance advantages, underscoring its practical applicability. Additionally, ablation studies highlight the synergistic effects of integrating FD-FE and ATCmix, emphasizing the critical role of their combined operation in enhancing classification performance.

CRedit authorship contribution statement

Xiuwei Chen: Writing – review & editing, Formal analysis, Conceptualization, Writing – original draft, Data curation. **Li Lai:** Supervision, Validation, Investigation. **Maokang Luo:** Visualization, Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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